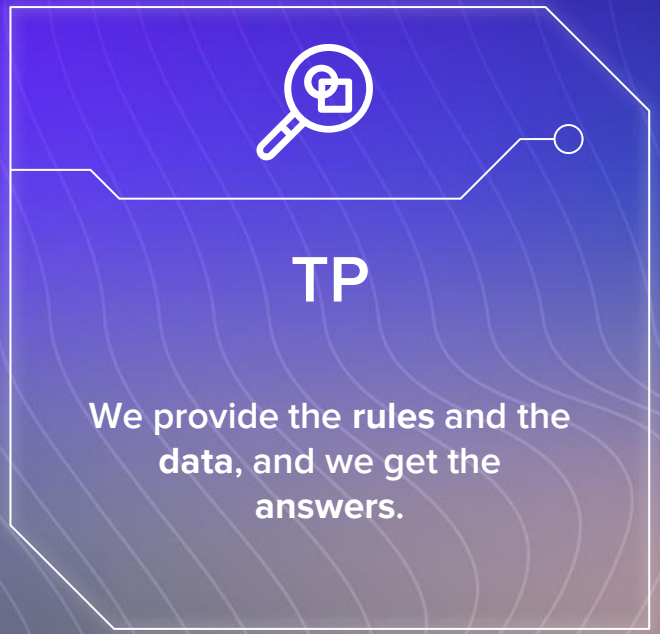
ML is our tool for solving complex, pattern-based problems.





Types of Learning

Supervised

Data and Labels

Philosophy: Training with labeled data.

Analogy: Teaching a child with flashcards (“This is an apple, this is an orange”).

Application in Embedded AI: Our lifeline — motion detection, speech recognition, signal classification.

Unsupervised

Only data – No labels

Philosophy: Discovering patterns and hidden structures in **unlabeled data**.

Example: Clustering and automatic data grouping.

Application in Embedded AI: Anomaly detection and sensor behavior analysis.

Reinforcement

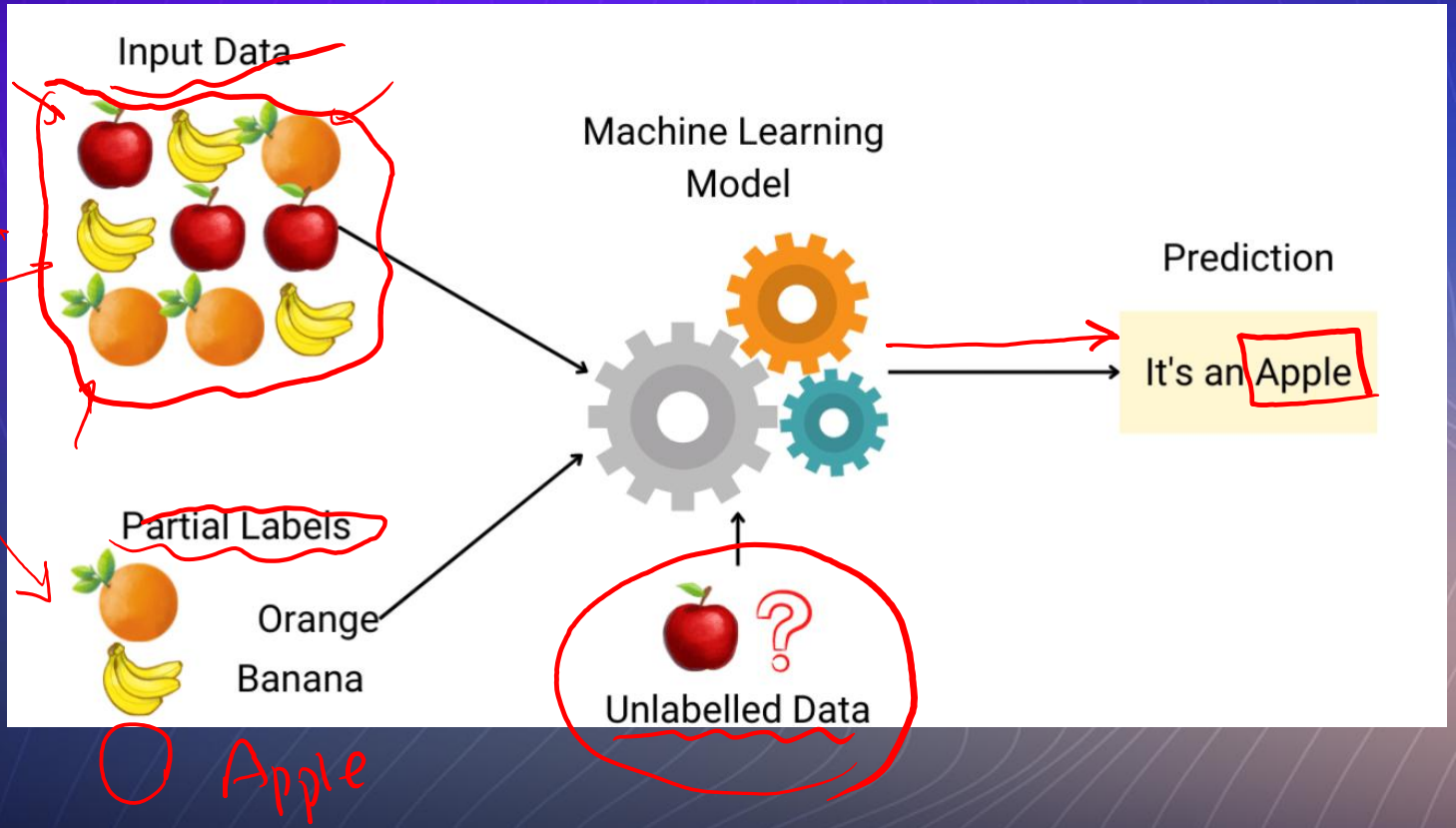
Trial and error (reward/penalty)

Philosophy: The **agent** learns through trial and error, receiving rewards or penalties.

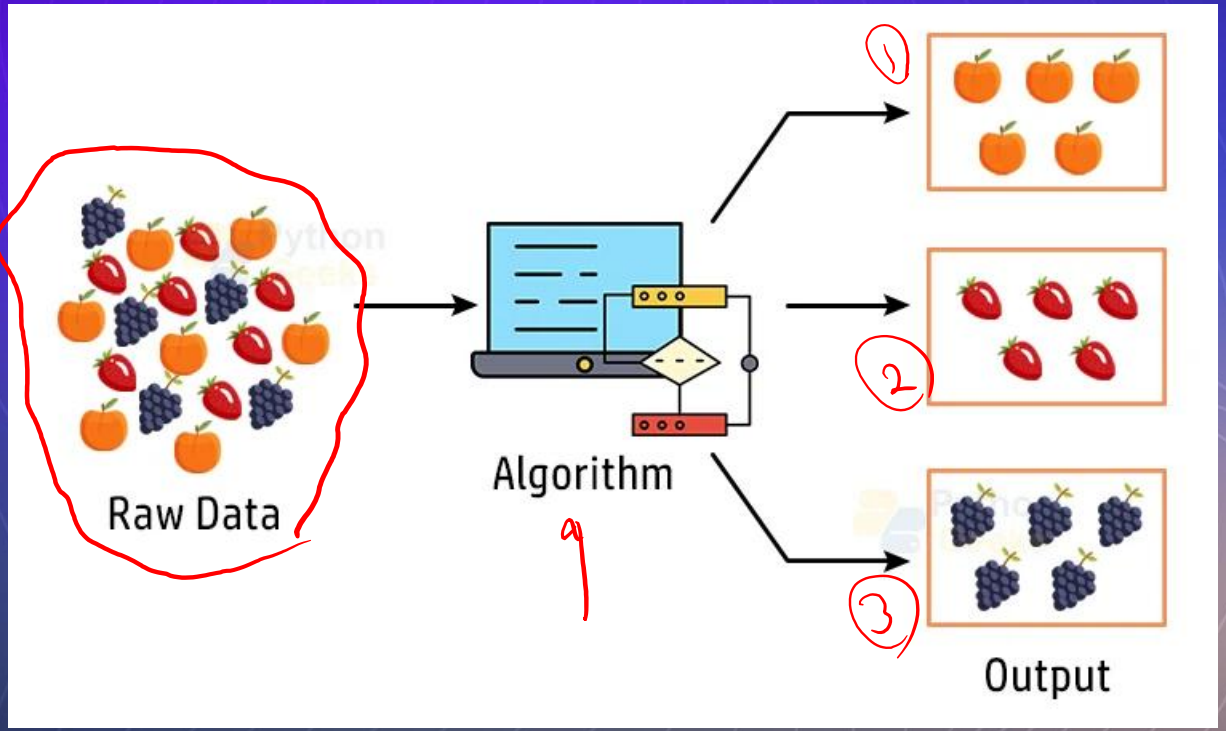
Analogy: Training animals or teaching computer games.

Application in Embedded AI: Used in advanced real-time control systems and robotics, but less common in beginner TinyML projects.

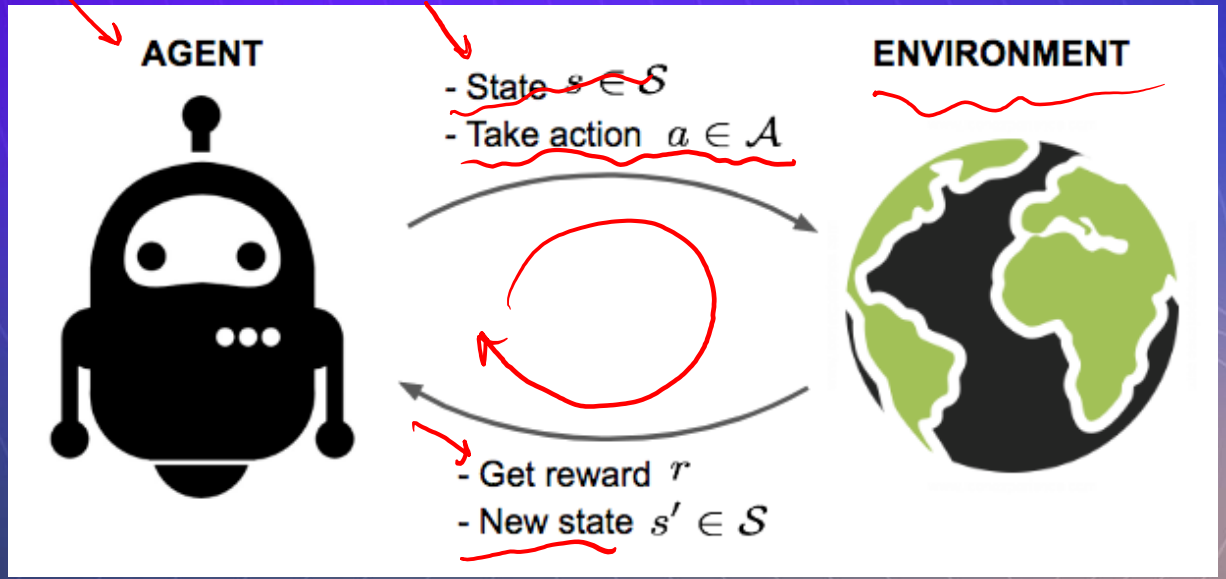
Supervised Learning - Classification



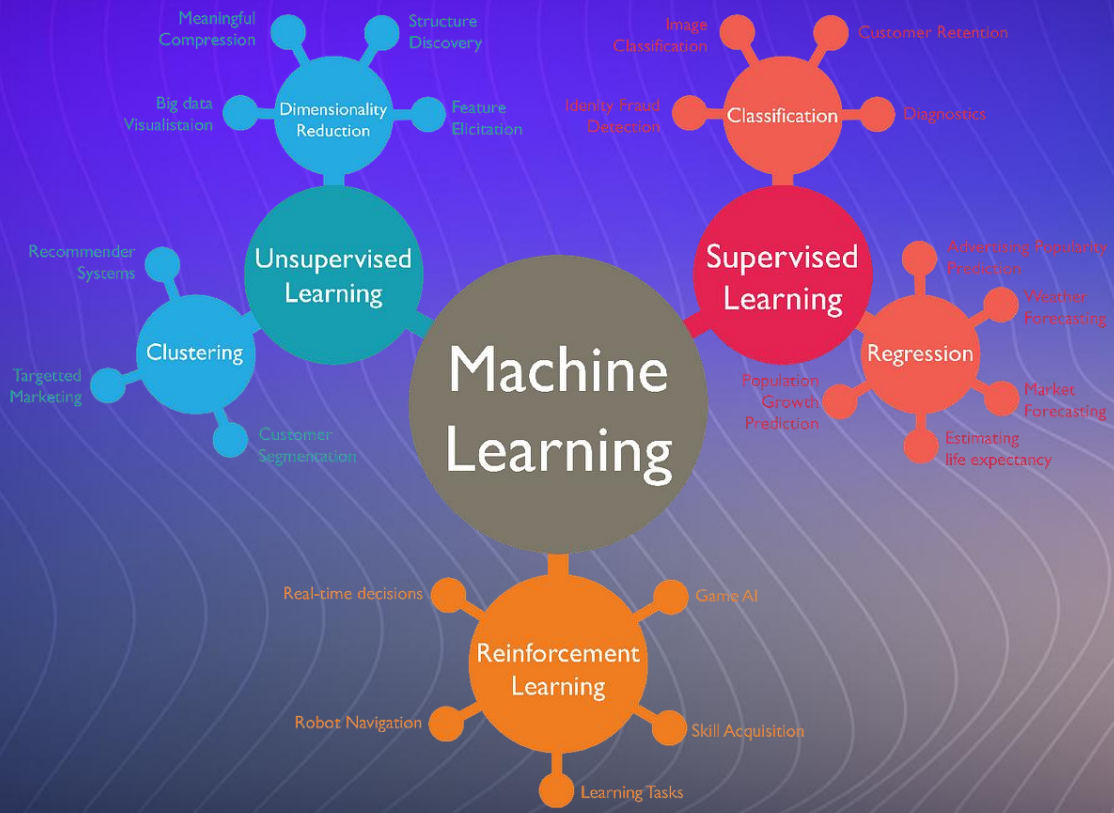
Unsupervised Learning - Clustering



Reinforcement Learning

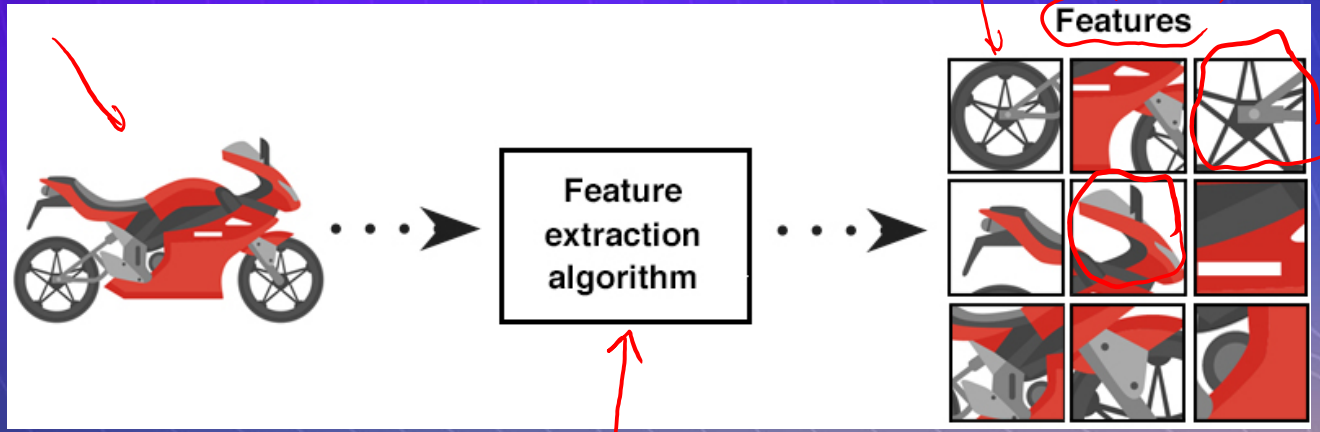


Reinforcement Learning



How algorithem works

Feature Extraction



Definition (Feature Engineering): Transforming raw data into meaningful inputs for the model.
In time series data: Instead of sending 1000 raw data points, send a summarized version.
Example: Convert 1 second of accelerometer data into 3 features — mean, standard deviation, and maximum acceleration.

How algorithem works

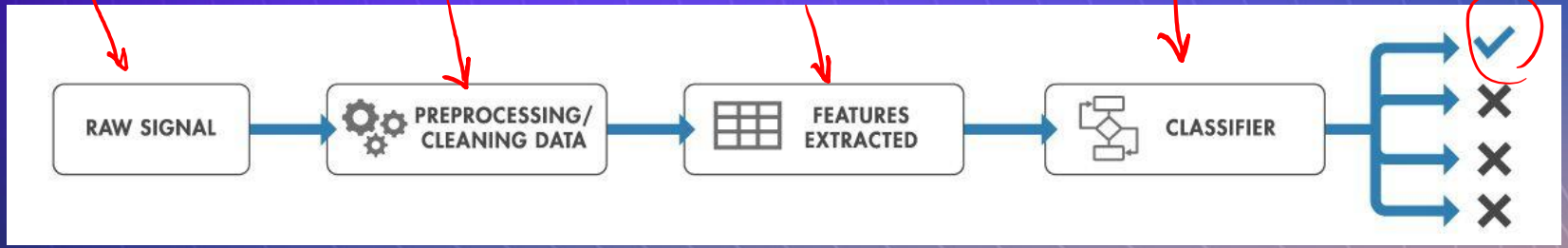
Preprocessing



Cleaning: Removing **outliers** and erroneous data.
Normalization / Scaling. Bringing all feature ranges to the same scale (e.g., between 0 and 1).
Importance: Prevents the model from focusing on features with larger numeric ranges.

How algorithem works

Classification Process



Types of Data



Statics

Constant over time, unchanging (e.g., images, static Excel tables).



Time Series

Time-dependent, continuous (e.g., stock prices, server temperature changes).

Data in EAI

Data sources: Sensors (IMU, microphone, temperature) rather than large images or text.

Challenge: High data volume in a short time span (noise) and high sampling frequency.

Importance of framing: Segmenting continuous data into meaningful chunks for the model.



Example of EAI data



IMU

(accelerometer and gyroscope) for motion and orientation

Microphone

For audio recording and storage



Types of DataSet

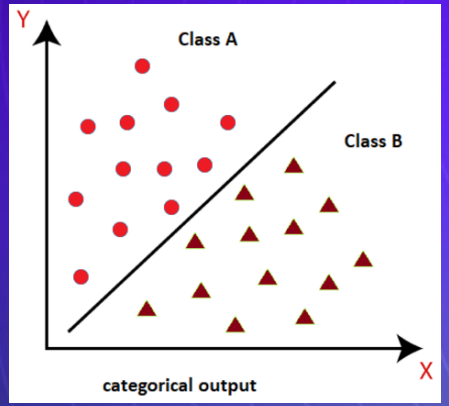
1 Structured Data

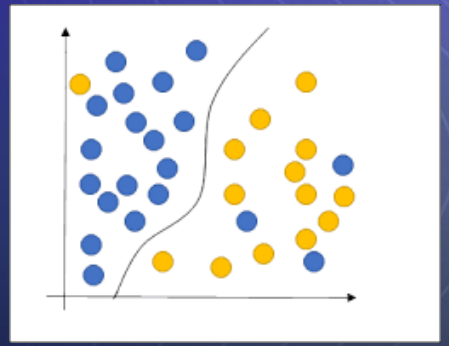
Organized data in rows and columns (e.g., tables, databases).

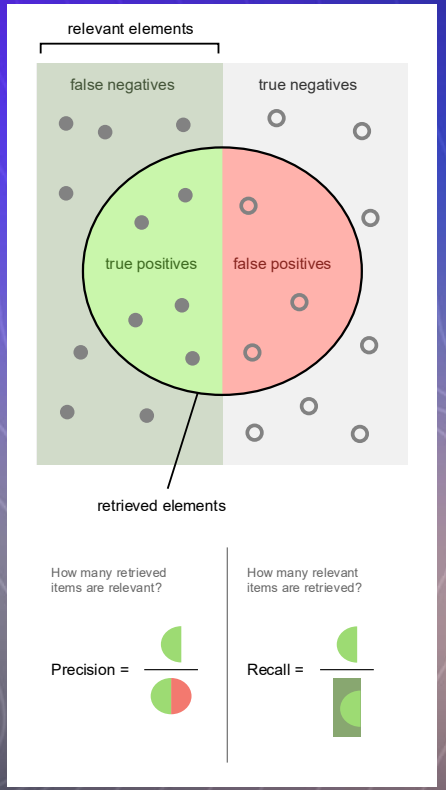
1 Unstructured Data

Raw, unorganized data (e.g., images, raw audio, text).

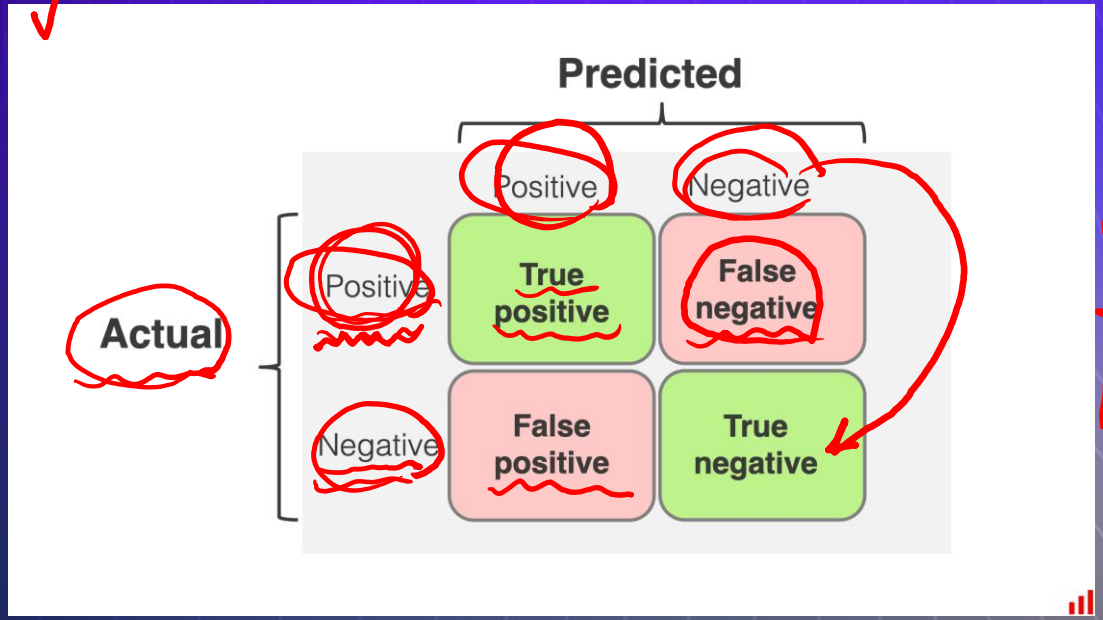
Train vs Test data







Performance Metrics



$$\textit{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\textit{Recall} = \frac{TP}{TP + FN} \text{ or } \frac{\text{True Positive}}{\text{Actual Results}}$$

$$\textit{Precision} = \frac{TP}{TP + FP} \text{ or } \frac{\text{True Positive}}{\text{Predictive Results}}$$

$$\textit{F - score} = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}}$$

F1-Score

Our tools

Scikit-learn

A collection of ready-to-use algorithms and tools for machine learning (classification, regression, clustering, etc.).

NumPy

A library for high-performance numerical computations and working with arrays and matrices.

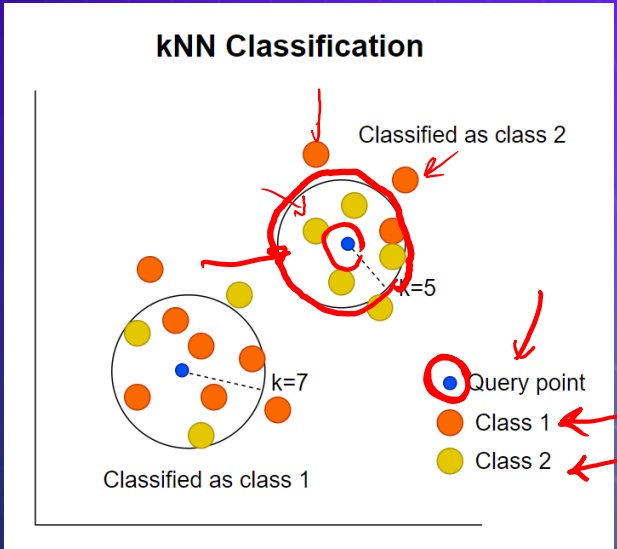
Pandas

A powerful tool for managing, cleaning, and analyzing tabular data (DataFrames).

MatplotLib

A library for creating various types of charts and data visualizations.

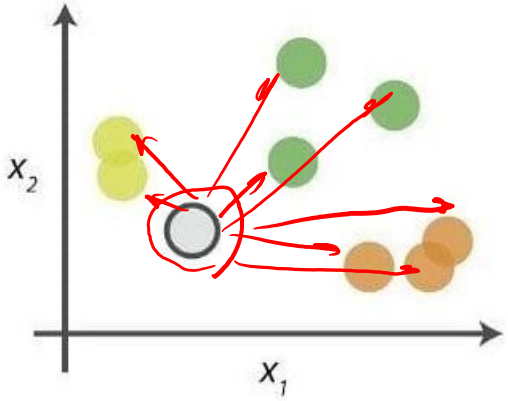
KNN



Goal: Build the first practical classification model.
Philosophy: A **lazy learning** and **distance-based** algorithm.
Core Logic: A new data point belongs to the class that most of its **K nearest neighbors** belong to.

KNN

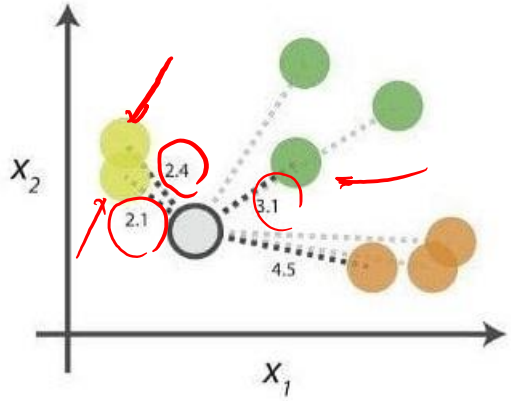
0. Look at the data



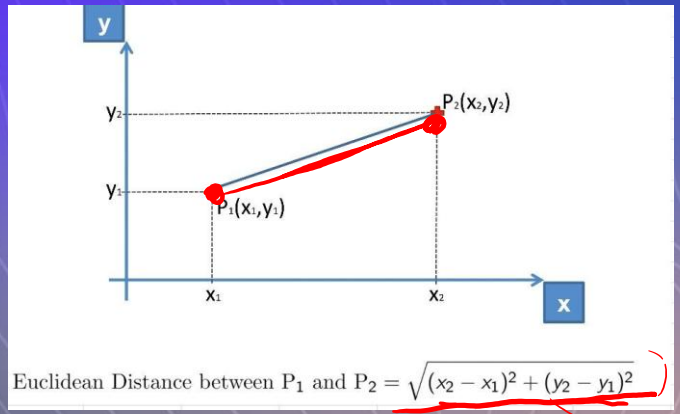
Say you want to classify the grey point into a class. Here, there are three potential classes - lime green, green and orange.

KNN

1. Calculate distances

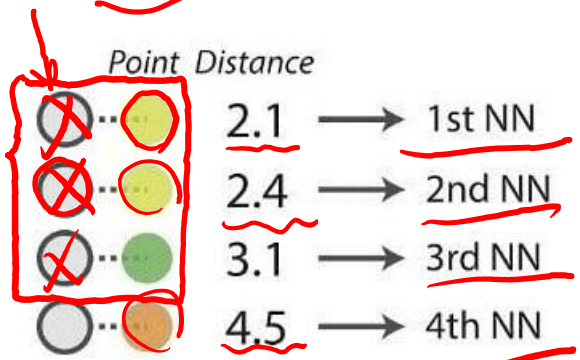


Start by calculating the distances between the grey point and all other points.



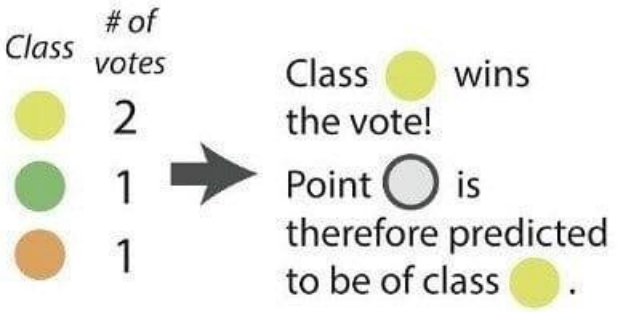
KNN

2. Find neighbours



Next, find the nearest neighbours by ranking points by increasing distance. The nearest neighbours (NNs) of the grey point are the ones closest in dataspace.

3. Vote on labels

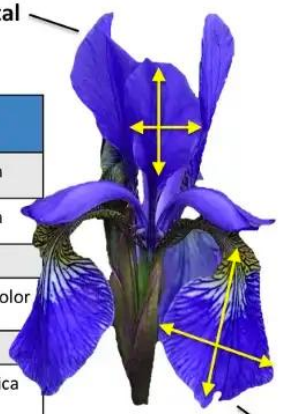


Vote on the predicted class labels based on the classes of the k nearest neighbours. Here, the labels were predicted based on the k=3 nearest neighbours.

Iris Dataset

Samples
(instances, observations)

	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica



Features
(attributes, measurements, dimensions)

Class labels
(targets)

Missing Value



Remove rows or columns that contain missing values

Deletion



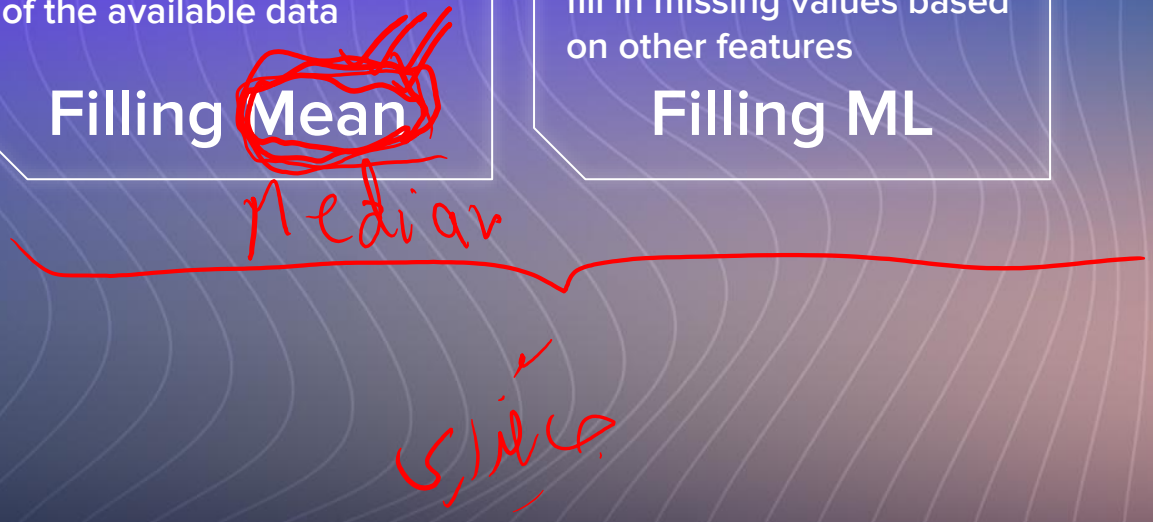
Replace missing values with the mean or median of the available data

Filling Mean



Use machine learning algorithms to predict and fill in missing values based on other features

Filling ML



Normalization

Standard Scaler

Transforms data to have a mean of 0 and a standard deviation of 1.
Useful when data follow a normal (Gaussian) distribution

$$z = \frac{x - \mu}{\sigma}$$

Min/Max Scaler

Rescales data to a fixed range, typically $[0, 1]$, Maintains the shape of the original distribution but compresses the scale

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Regression

Linear Regression

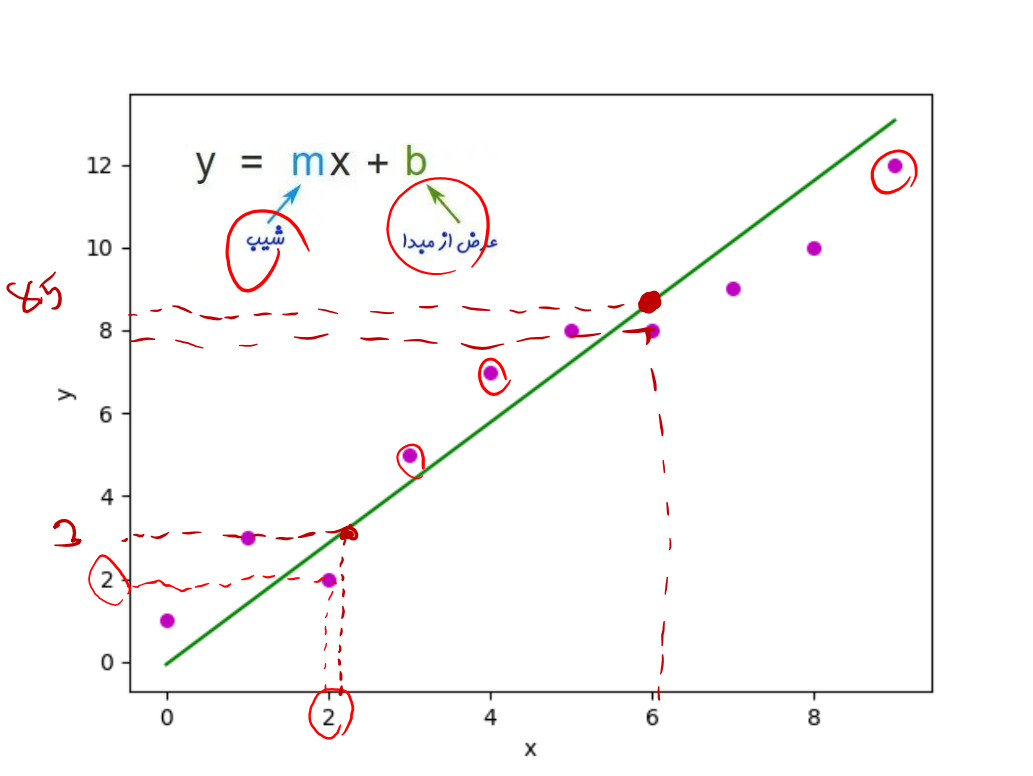
Predicts continuous values (regression) using a straight line.
Fits a linear relationship between input features and output; ideal for trend prediction (e.g., temperature, price).

Logistic Regression

A linear model used for binary classification (two classes).
Uses the logistic (sigmoid) function to output probabilities — e.g., “yes/no” or “on/off” decisions.

Perfect for microcontrollers and low-power AI applications where memory and speed are limited.

Simple Linear Regression



Linear Regression Equation

$$Y = a + bx$$

$$a = \frac{[(\Sigma y)(\Sigma x^2) - (\Sigma y)(\Sigma xy)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

$$b = \frac{[n(\Sigma xy) - (\Sigma x)(\Sigma y)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

Multiple Linear Regression

